

Invisible Coordination Work: Field Stations as Scientific Caregivers

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ABSTRACT

Field stations are an important part of the global scientific research infrastructure providing access and logistical support that makes scientific contributions across domains from ecology to archeology possible. There are over 1000 field stations worldwide. However, little attention has been given to the cooperative work at field stations or how computer-supported cooperative work could benefit this cooperation. We conducted six ethnographic interviews from August 2022 through October 2022 with staff representing six field stations. The interviews made visible the articulation and coordination work needed for *scientific caregiving* that field stations do before, during, and after research teams do research at the site. This poster represents a portion of preliminary work in a larger study on cooperative work at field stations.

CCS CONCEPTS

• **Applied computing** → **Environmental sciences**; • **Human-centered computing** → **Collaborative and social computing**.

KEYWORDS

biological field stations; coordination work; scientific caregiving

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1 INTRODUCTION

For field scientists, like ecologists, archeologists, and geologists, the natural environment is not just a space they go to, but it is a place they actively work in [6]. Field stations and marine labs were established as early as the 1870s, not long after traditional laboratories, to support field scientists with logistical support, easier access to the field, and laboratory infrastructure. Field stations expanded the type of science that could be done and lent credibility to the research. By 1940, 48 stations were built in inland North America. Today there are over 1000 stations worldwide [15].

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Field stations and their infrastructure—human, physical, and digital—are scientific caregivers dedicated to caring for the place and the field scientists that visit. Care is defined by Fisher and Tronto [5] as:

“everything that we do to maintain, continue and repair “our world” so that we can live in it as well as possible. That world includes our bodies, ourselves, and our environment, all that we seek to inter- weave in a complex, life-sustaining web.”

There are four phases of caring which include: (1) caring about; (2) taking care of; (3) caregiving, and (4) care-receiving [5, 14]. Our research shows that field stations’ care practices are sites of cooperative work, where field stations and researchers must do additional articulation work [10] extraneous to their goal of producing scientific results in the form of caregiving and care receiving. Prior work has studied cooperation across large National Science Foundation (NSF) projects like the Long Term Ecological Research (LTER) Network and cyberinfrastructure projects like the Ocean Observatories Initiative (OOI) [8, 9]. Prior studies have also focused on the work practices of scientists and their cooperation with computer scientists [3, 11]. However, little attention has been given to the coordination work at field stations, particularly from the perspective of field stations or how computer-supported cooperative work could benefit this cooperation [1].

2 METHODS

The design of this study and focus on biological field stations was informed by the first author’s ongoing work as a data management consultant collaborating with a field station director who oversees the management of two active field stations. However, the empirical data collected for this paper comes from six semi-structured ethnographic interviews conducted from August 2022 through October 2022 that represent six field stations independent of the consulting work. Collectively, the six field stations represent a variety of dimensions, including financial structure (independent or university-affiliated), operation season (annual or seasonally), physical infrastructure (rustic to state of the art), and mission (research, teaching, or public outreach).

We gained access to the Organization for Biological Field Station (OBFS) community through a prior, unpublished study on how field stations support permitting and declarations requirements needed by researchers accessing field sites. Participants responded to an email to the OBFS general list-serve and self-selected. All participants and the stations that they worked at were anonymized.

The interview protocol included questions about the history of the field station, how the participant arrived at the station, how

research teams access the site, the relationship between the field station and the visiting scientists, and the expectations of sharing data, results, or papers back to the field station. Each semi-structured interview lasted about one hour via Zoom, recorded, and then transcribed by Otter.ai. The transcripts were reviewed and corrected.

Our interview analysis followed the Grounded Theory Method [4]. Using atlas.ti, we created open codes line-by-line for a single interview to develop lower-level concepts. Codes were then grouped to create categories. Subsequent interviews were analyzed following this approach of open coding building to categories. Constant comparison of the categories was done as new data were added. Over 100 memos were created throughout the entire analysis. In total, over 2000 open codes were created across the six interviews, and codes were consolidated into 37 categories. Both authors reviewed the initial analysis, which led to the definition of the initial core category of *scientific caregiving*.

3 PRELIMINARY FINDINGS AND DISCUSSION

Field stations coordinate with experimental, hypothesis-driven research and the teams that do the work over the life of research projects. To support this work, field stations act as *scientific caregivers*. By this, we mean that they both provide physical and emotional care and support to the researcher, across their career stages from students to faculty, and they also must care for the field station infrastructure and the place in which the field station is located. The primary functions of field stations, by necessity, serve a variety of roles that could be likened to the work of a car rental business, laboratory facility, hotel, conference center, classroom, and dive shop, among many others. Several participants expressed the diversity of jobs or activities needed, from neighbor management to janitor to construction worker to tour guide and, if there is time, scientist. This caregiver role sometimes relegates the sustained science at field stations, often as long-term, place-based observations, to a secondary status pursued in gaps and with limited funding.

3.1 Scientific Caregiving for Research

Field stations take an active support role in the planning and data collection phases of the research lifecycle when research teams are trying to gain access to the site and while on-site at the field station. P4 said:

"The whole point of our existence is to facilitate research."

Field stations navigate locally or federally protected land through relationships with federal and state agencies. Often teams choose to work with field stations because it is easier to do their research projects on this land instead of working with state or federal partners. P2 said:

"We enable and approve research to happen. If someone wants to experiment with carbon and create a dome over some trees, the field station can facilitate it, but that research would not be allowed on the adjacent public land for other reasons."

Each station has a different process for applications to work at the field stations. Generally, these are brief proposals that describe

the science activity, methods to be used, the needed station infrastructure, like housing and equipment, and if the work is planned in sensitive areas. Staff review research projects through bespoke online application systems or email and attempt to make sure that the work is scientifically sound and doesn't conflict with existing research or harm the place for future work. P4 said:

"You know, but often, a lot of that [application coordination] is done offline, meaning outside of [bespoke application system], other communications, either email or phone calls."

Visiting scientists expect to show up and have everything they need, so scientific field station staff spend most of their time and effort caring for, i.e., hosting, visitors. This hosting includes providing access to housing, equipment, and laboratory space. Stations also help teams liaise with the community, navigate permit processes for collecting restricted materials, procure supplies, or find medical care in a new place. Stations go to great lengths to provide a productive workspace for the research teams. P5 said:

"So we've bent over backward in many ways to accommodate people from elsewhere who want to come and do research."

In *The Ecology of Place: Contributions of Place-Based Research to Ecological Understanding*, Ian Billick, Director of the Rocky Mountain Biological (RMBL), a field Station in Gothic, CO, describes, "one of the joys and benefits of working at the field station is that of sharing observations with other scientists." This is ephemeral, and for field research, it is expensive and not repeatable [2]. In addition, teams rotate through the field station, staying a few days to several months. Most of their analytical work is done at their home institutions. It surprised us to learn of the lag between fieldwork and analysis. The field station may never hear from the team again unless they want to return for subsequent data collection activities. P1 overcame this ephemeral, serendipitous challenge through long-term staff:

"And having a really stable person or a stable leadership on site, oftentimes was the connection from a past project to a future one. And yeah, not everything was well documented. I often teased [our long-term staff person] that I just wanted to kind of insert a computer cable in the back of [their] head when [they were] retiring sort of thing. I would go to [their] office once a week, once a month, and ask a question on this, 'can you help me? can you point me in the right direction?' And [they] would tell me or show me."

P1 went on to describe the importance of these relationships and the care that the long-term staff person took with all people saying:

"And [the long-term staff person] was a very unique person, and that [they] really didn't care if you were a distinguished professor, or you were a new masters student. If you had a technique, or you had a tool, or you had an idea that [they] saw some value in sort of thing, [they] would connect you to a dataset, or [they] would put you in contact with one of [their] staff, or [they] would collaborate with you on a publication to be able to kind of build that relationship or growth."

And that really got people off to a great start. But I think the connection between projects is between, you know, the measurements, or resampling is a very person or people-related sort of thing. And that's hard to facilitate with a website or a dataset, or a project."

Yet, when this long-term staff member retired and ultimately passed away, essential but undocumented knowledge was lost. One area in which we see field stations increasing capacity is the support of the on-site data management needs of research teams because curating scientifically significant data about the place helps future scientists better understand it [13]. This leads to the other significant activity at field stations, that of *scientific caregiving for the place*.

3.2 Scientific Caregiving for the Place

Field stations spend a great deal of time constructing the place of the field station. Field station staff tell stories to visitors about the founding of the station, the evolution of the infrastructure, the relationships that the station has with other institutions and the local community, and key features of the ecosystem. This social construction of place, or "ecology of place," is what attracts researchers to come and work at these stations. After research teams submit their applications to visit, field stations do significant cooperative and articulation work to avoid research conflicts, particularly around access to sensitive areas. Sensitive areas vary from station to station but include historically sensitive sites, like burial plots, that may not be obvious. They also may include scientifically significant sites where other studies are ongoing or where precautions are needed to avoid spreading an invasive species. While most requests are accepted, for those that are not, field station staff point the requester to other types of land that might be more suitable. P2 noted that the cooperation doesn't just span the present but also considers the future:

"The details of what the research team wants to do have to be approved by our institution. So if, for example, a researcher proposes to dump chemicals in a stream and see what happens to the biota, we would review that and say, well, is that consistent with the kinds of places that we're working with? And does that impact future researchers' ability to understand that part of the system?"

Interview participants believed in the importance of a deep understanding of the place. In addition to managing access to the physical place, field stations have staff scientists who create a bridge to visiting researchers' hypothesis-driven experiments by collecting contextual, long-term data like weather, soil moisture, alligator size, or streamflow data. Alone, this contextual data is often seen as uninteresting and unworthy of journal publication. P6 noted the value of long-term data increasing in conjunction with data collected by visiting research teams or when unanticipated events occur:

"These long-term data programs don't create as many publications, because it's not like you're going to publish two years of data at a time on some of these things. The long-term benefit is still big. Because when those unusual occurrences happen, you have this baseline, and other people want to study things on top of that."

These examples of the integration of field station collected and visiting research collected data lead to what Billick describes as the collective, social construction of place as the 'ecology of place.' The ecology of place vision is to integrate all the data, information, and knowledge produced at the field station [2]. The CSCW to support the social construction of place is lacking in part due to the invisibility of field station caregiving.

3.3 Invisible Caregiving and Dichotomy of Spheres

The primary focus of scientific caregiving done by field stations is almost entirely in service of hypothesis-driven research outcomes like publications and published datasets. Yet we found a curious and problematic disconnect. Despite the significant work done by the field station, place-based scientific results often do not mention the field station in consistent or standard ways, and sometimes the station is not mentioned at all. The lack of connection and visibility between the research outputs created by the visiting teams and the field stations shows that the scientific caregiver work is "shadow work" or "expertise is hidden from view" [12]. It also might be considered a breakdown in cooperation between the caregiver and the researcher as the care receiver [14]. This disconnect highlights the separate spheres dichotomy between research and field stations. While the dichotomy of spheres normally denotes a traditional gendered division of labor, we argue that the field stations' scientific caregiving work exists in the private domestic sphere of hosting, organizing, and supporting others [7, 14]. Field research, on the other hand, is highly visible and exists in the public sphere producing scientific papers and advancing reputations. Field stations try to overcome this disconnect through manual, labor-intensive processes of bringing the research results back to place. Some stations have annual reporting requirements for which they email all visitors from the past year and ask for publications or published datasets. That generally does not yield much response. Without research teams providing results of their work, it is difficult for the field stations to make the case for the value they provide to the broader community. P5 described their approach by saying:

"Well, first of all, we have someone who spends some time searching up publications that mentioned our site to make sure that they acknowledged us because NSF has this system called the NSF Public Access Repository (NSF-PAR) where they want you to report publications funded by a certain grant. Okay, so our site is supposed to put all of our publications on it. But that requires explicit mention of [the site]. Or better yet, the grant number."

P5 went on to describe an additional, unusual step of asking researchers to submit corrections to journals, saying:

"And so occasionally, we find a paper that clearly used data from our site, usually by someone who had a fleeting presence or took samples from the archive or whatever and disappeared. And then years later, they published a paper and said, 'samples were taken from the [the site], you know, but they didn't acknowledge [our funding]. And in those cases we've actually asked them to let us publish an erratum basically, that gives

the proper funding information. And then we can list the paper in NSF PAR."

Finally, P5 emphasized that the update isn't just for vanity or to claim credit to their funders. It serves an essential research purpose:

"and more importantly, it makes it more discoverable. We don't want potentially important studies using our samples to go under the radar and never be seen by people looking for the papers. And then, we put that along with other papers on our website, whether it has the explicit mention or not, but it goes into the NSF system if it has the explicit mention of the grant funding."

While many field stations do not go through the elaborate process described by P5, many have a curated Zotero bibliography on their website. These bibliographies are valuable to the field station and future research by connecting the scientific results from field research back to the place. In addition, P5 and a few other stations have data managers on staff or have allocated staff development to increase data stewardship skills to provide access to the digital place. We found that these field stations leverage cooperative relationships with cross-station organizations like OBFS or the Environmental Data Initiative (EDI) to come to a consensus on best practices for managing data. Field station staff discussed using general data repositories like Dryad and Figshare as options or local computer storage to support saving smaller datasets. This scientific caregiving work of reconnecting the results to the place repairs the disconnect and makes the invisible visible. Through this care, it allows present and future research teams to build on prior work. Two initial speculative reasons for this dichotomy of spheres are that the analysis and the papers often happen months to years after the field visit. Researchers move on, and the field station's importance fades. There is also a mismatch between place-based and phenomena or species-based work. Research teams may be working across several sites, so the importance of place is again lost or seemingly less important relative to the study species.

4 CONCLUSION AND NEXT STEPS

When things work well at field stations in support of research, they have been considered. This initial study of just six sites surfaced the enormous amount of articulation and coordination work performed by field stations to care for the research and the place, as well as the minimal computer support currently used. The study showed that the burden lies on the field stations to carry most of the cooperative and articulation caregiver work needed to support visiting research teams. Future participatory action research will work with field station partners to develop a collaborative infrastructure that makes visible the long-term connections between research teams, their projects, and the place to reduce the burden on the field stations. In addition, we will support field stations as they teach research teams how to cooperate more efficiently with field stations and each other. Finally, field stations are a cooperation nexus. We expect to expand this work to look at other types of cooperation at field stations, including with the local community, state and federal agencies, and hosting institutions.

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